ModelIt!

Student Activity Pack

How Does the Brain Create Consciousness?

Modeling Neural Correlates, Emergence, and the Architecture of Subjective Experience

Grade 12L3 | NGSS HS-LS1-2 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Close your eyes for 10 seconds and pay attention to your own awareness. What is it like to be conscious? Can you describe the experience of experience itself?**

2. **When you dream, you have vivid conscious experiences without any input from the outside world. What does this tell you about how the brain generates consciousness?**

3. **If a brain-injured patient has their eyes open but shows no response to commands, how could you determine whether they are conscious inside but unable to communicate?**

4. **What is the difference between being awake and being conscious? Can you be one without the other?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Sensory Cortex Activation
- Arousal State
- External Stimulus Complexity
- Thalamo-Cortical Loop Activity
- Prefrontal Integration Level
- Gamma Synchrony Index
- Default Mode Network Activity
- Information Integration (Phi)
- Recurrent Processing Depth
- Conscious Experience Index

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Full Awareness

Arousal: Alert | Stimuli: Complex | All systems active

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Anesthesia

Arousal: Minimal | Thalamo-cortical loops: Disrupted | Sensory cortex: Partially active

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Dreaming

Arousal: Moderate (REM) | External stimuli: None | Default Mode: High | Recurrent Processing: Active

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Interoceptive Processing — The brain's representation of internal body states — heartbeat, breathing, gut feelings, temperature — that contribute to the embodied sense of self and emotional consciousness, proposed by Antonio Damasio as essential for the feeling of being a conscious organism rather than a disembodied information processor

Attention Network Control — The dorsal and ventral attention networks that select which information enters conscious awareness — attention is the gatekeeper of consciousness, and understanding how attention selects certain stimuli for conscious processing while filtering others reveals the bottleneck in the stream of consciousness

Neurochemical Modulation — The influence of neurotransmitters (serotonin, dopamine, norepinephrine, acetylcholine) on conscious experience — psychedelic drugs that alter serotonin signaling produce dramatic changes in consciousness, suggesting that neurochemistry shapes not just the content but the fundamental character of experience

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

You are aware that you exist. You experience the redness of red, the pain of a stubbed toe, the joy of music. But 86 billion neurons firing electrochemical signals should not produce subjective experience any more than 86 billion transistors should produce feelings. How does the physical brain create the non-physical experience of consciousness? Use evidence from your simulation to support your answers.

1. **How does your model explain why anesthesia eliminates consciousness even though sensory cortices continue to process information?**

2. **What does the dreaming scenario reveal about the brain's ability to generate conscious experience internally, and what neural components make this possible?**

3. **Why might the neural correlates of consciousness be easier to study than the neural causes of consciousness, and what is the scientific significance of this distinction?**

4. **How would adding interoceptive processing or neurochemical modulation to your model change the predicted Conscious Experience Index, and what does this suggest about embodied consciousness?**

5. **What are the limitations of modeling consciousness with neural components when the hard problem suggests that subjective experience may not be reducible to physical processes?**

8 STEM Challenge: Design a Consciousness Assessment Protocol for Brain-Injured Patients

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YOUR ENGINEERING MISSION

Design a clinical protocol using neuroscience principles from your model to assess the level of consciousness in brain-injured patients who cannot communicate, distinguishing between vegetative state, minimally conscious state, and covert consciousness.

THE SCENARIO

A patient has suffered a severe traumatic brain injury and appears to be in a vegetative state — eyes open but no purposeful behavior. However, research has shown that up to 15-20% of apparently vegetative patients show signs of covert consciousness when tested with advanced neuroimaging. Your neuroethics team must design an assessment protocol using brain imaging and stimulation techniques to determine whether this patient has hidden awareness, and develop ethical guidelines for treatment decisions based on the results.

Engineering Design Thinking

1. WHAT SPECIFIC NEUROIMAGING TECHNIQUES DOES YOUR PROTOCOL USE AND WHAT NEURAL SIGNATURES DOES EACH ONE DETECT?

2. HOW DOES YOUR PROTOCOL DISTINGUISH BETWEEN REFLEXIVE BRAIN RESPONSES (UNCONSCIOUS) AND GENUINE CONSCIOUS PROCESSING?

3. WHAT IS THE SENSITIVITY AND SPECIFICITY TRADE-OFF IN YOUR PROTOCOL — IS IT BETTER TO FALSELY DETECT CONSCIOUSNESS OR FALSELY MISS IT?

4. WHAT ETHICAL FRAMEWORK GUIDES TREATMENT DECISIONS WHEN THE ASSESSMENT IS UNCERTAIN?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Neuroscientists who study consciousness at research universities and institutes earn $70,000-$150,000/year. Clinical neurologists who assess and treat disorders of consciousness earn $200,000-$350,000/year. Neuroimaging researchers who develop brain scanning techniques earn $75,000-$140,000/year. Neuroethicists who advise on moral questions about brain science earn $65,000-$130,000/year.